Dynamical Systems: Theory to Applications

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Introduction to Dynamical Systems

Dynamical Systems

It is an abstract system consisting of :

1. a space (state space or phase space).
2. a mathematical rule describing the evolution of points in that space.

Classification

- Classified as Discrete & Continuous dynamical systems according to the time parameter is discrete or continuous.
- Continuous dynamical systems are classified as Autonomous & Non-autonomous according to the explicit independence or dependence of their equation of motions on the time parameter.

Autonomous Systems

Definition

Let $\mathbf{x} = (x_1(t), x_2(t), ..., x_n(t)) \in X \subseteq \mathbb{R}^n$. An autonomous dynamical system is written in the form,

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}),$$

Where $\mathbf{f} : X \to X$, dot denotes differentiation with respect to the time parameter t, $\mathbf{f}$ is viewed as a vector field in $\mathbb{R}^n$. $\mathbf{f}$ is sufficiently smooth.

Example

A simple example of autonomous system is,

$$\frac{dx}{dt} = 3x + 4y,$$
$$\frac{dy}{dt} = 7x + 9y.$$

Key Terms

The key elements for studying dynamical systems are:
1. Phase space. (Where the flow of the phase point is studied)
2. Phase portrait. (The plots in the space)
3. Equilibrium Points. (Where the system does not exhibit any change)
4. Stability and bifurcations. (The states and their changes in the system)

Linear Stability Theory

Linear Stability Theory

The basic idea of linear stability theory is to linearize the system near a fixed point to understand the dynamics of the entire system near that point.

Stability Analysis for Autonomous Systems

Consider a two-dimensional dynamical system,

$$\begin{aligned} \dot{x} &= P(x, y), \\ \dot{y} &= Q(x, y), \quad (\,\dot{}\equiv d/dt\,) \end{aligned}$$

Where P and Q have continuous first partial derivatives for all x and y and they does not depend explicitly on the variable t.

Stability Analysis (Continued)

Equilibrium Points

Let $X = X^* = X_0 \equiv (x_0, y_0)$ be the equilibrium point of the above system. Then, $\dot{x} = \dot{y} = 0$ at (x_0, y_0). So, $P(x_0, y_0) = Q(x_0, y_0) = 0$.

Determination of Stability

In order to determine stability, we slightly disturb it (or we say that we infinitesimally perturb it),

$$x = x_0 + \xi(t),$$
$$y = y_0 + \eta(t),$$

where ξ, η are components of a small disturbance from the fixed point. Now to determine the resultant motion, we use Taylor series expansion of the functions about the equilibrium point,

$$P(x, y) = P(x_0 + \xi, y_0 + \eta) = P(x_0, y_0) + \frac{\partial P}{\partial x}\Big|_{(x_0,y_0)} . \ \xi + \frac{\partial P}{\partial y}\Big|_{(x_0,y_0)} . \ \eta + \text{higher order terms in } (\xi, \eta).$$

$$Q(x, y) = Q(x_0 + \xi, y_0 + \eta) = Q(x_0, y_0) + \frac{\partial Q}{\partial x}\Big|_{(x_0,y_0)} . \ \xi + \frac{\partial Q}{\partial y}\Big|_{(x_0,y_0)} . \ \eta + \text{higher order terms in } (\xi, \eta).$$

Stability Analysis (Continued)

Neglecting the higher order terms (since they are tiny), the two-dimensional dynamical system can be rewritten as,

$$\dot{\xi} = \dot{x} = a\xi + b\eta,$$
$$\dot{\eta} = \dot{y} = c\xi + d\eta,$$

where a, b, c, d are given by,

$$a = \frac{\partial P}{\partial x}\Big|_{(x_0,y_0)}, \quad b = \frac{\partial P}{\partial y}\Big|_{(x_0,y_0)},$$
$$c = \frac{\partial Q}{\partial x}\Big|_{(x_0,y_0)}, \quad d = \frac{\partial Q}{\partial y}\Big|_{(x_0,y_0)}.$$

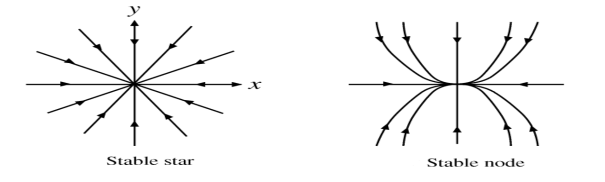
Therefore the system evolves to the matrix of the form,

$$\begin{pmatrix} \dot{\xi} \\ \dot{\eta} \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix}$$
$$\text{i.e. } \dot{X} = AX,$$

Where, A is the Jacobian matrix at the fixed point (x_0, y_0).

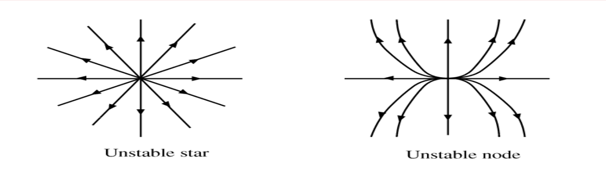
Eigen Values & Classification of Equilibrium Points

Case 1: $\lambda_1 \leq \lambda_2 < 0$



$\lambda_1 = \lambda_2$ $\qquad$ $\lambda_1 < \lambda_2$

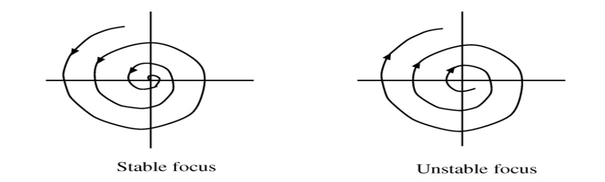
Case 2: $\lambda_1 \geq \lambda_2 > 0$



$\lambda_1 = \lambda_2$ $\qquad$ $\lambda_1 > \lambda_2$

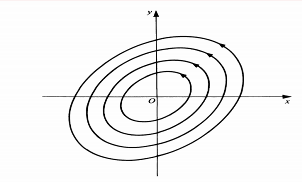
Eigen Values & Classification of Equilibrium Points(Contd.)

Case 3: λ_1, λ_2 complex conjugates



Real Part < 0 Real Part> 0

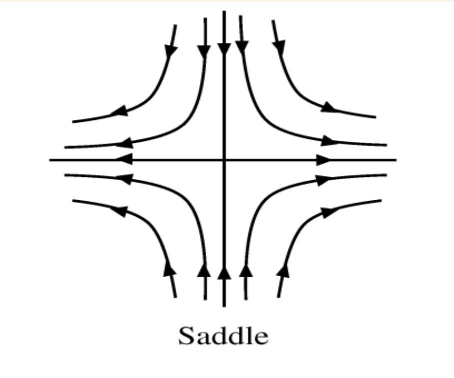
Case 4: λ_1 and λ_2 pure imaginary and complex conjugates



Center/Elliptic

Eigen Values & Classification of Equilibrium Points(Contd.)

Case 5: $\lambda_1 < 0 < \lambda_2$



Bifurcation Analysis

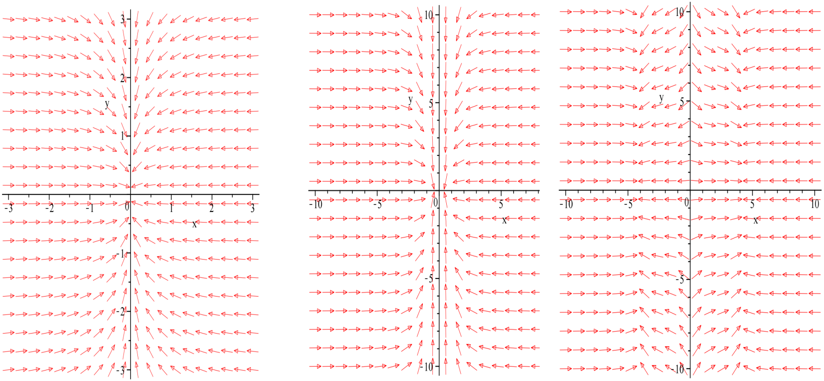
Bifurcation

Depending on the values of certain parameters, sudden qualitative changes occur in the systems under study. This is called Bifurcation.

Example

Consider the dynamical system,

$$\dot{x} = \mu x - x^3,$$
$$\dot{y} = -y, \quad \text{Where } \mu \text{ is a parameter.}$$



$\mu < 0$ $\mu = 0$ $\mu > 0$

Figure: Phase trajectories corresponding to different values of μ

Bifurcation Analysis (Contd.)

Bifurcation Diagram

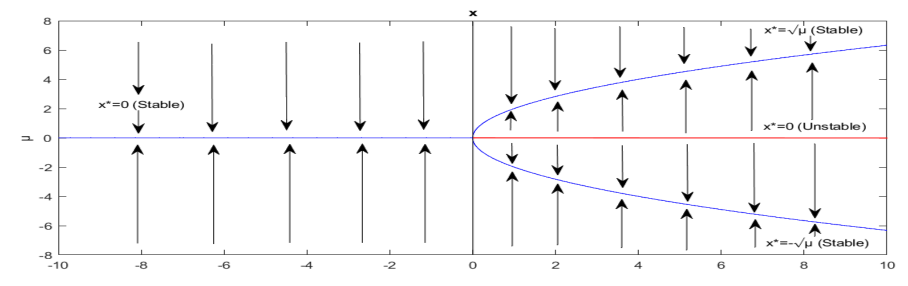


Figure: Bifurcation Diagram (arrows indicating the stable and unstable nature of equilibrium points)

Limitation of Linear Stability Theory

Linear Stability Theory cannot determine the stability of critical points with Jacobian having eigen values with zero real parts.

A Dynamical System Analysis of a Three Fluid Cosmological Model

Paper Details

- **Title of the Paper:** A Dynamical System Analysis of Three Fluid Cosmological Model.
- **Author(s):** Nilanjana Mahata and Subenoy Chakraborty. (Department of Mathematics, Jadavpur University, Kolkata-700032, West Bengal, India)
- AIP Conf. Proc., 2293, 290002 (2020)
- **Arxiv Number:** arXiv:1512.07017

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

Application of Dynamical Systems

To study the model, we have some basic equations. We have to transform them into autonomous dynamical systems as per our requirement to analyze the model properly.

Energy Density and Pressure

Dark energy is driven by a scalar field ϕ with potential $V(\phi)$ whose energy density and pressure is given by

$$\rho_d = \frac{1}{2}\epsilon\dot{\phi}^2 + V(\phi) \text{ and } p_d = \frac{1}{2}\epsilon\dot{\phi}^2 - V(\phi).$$

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

Klein-Gordan Equation

Klein-Gordan equation for the scalar field is

$$\ddot{\phi} + 3H\dot{\phi} + \frac{dV}{d\phi} = 0$$

Einstein Field Equations

The Einstein field equations can be written as (here $k = 8\pi G$ is gravitational constant, $c = 1$)

$$3H^2 = k(\rho_m + \rho_d + \rho_b),$$

$$2\dot{H} = -\frac{k}{2}(\rho_m + \rho_b + p_b + \dot{\phi}^2),$$

where H is the Hubble parameter.

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

System of Equations

From the previous equations, we obtain the system of equations as

$$3H^2 = k(\frac{A}{a^3} + \frac{1}{2}\dot{\phi}^2 + V(\phi) + Ca^{-3}),$$
$$2\dot{H} = -\frac{k}{2}(Aa^{-3} + \nu Ca^{-3\nu} + \dot{\phi}^2),$$
$$\ddot{\phi} + 3H\dot{\phi} + \frac{dV}{d\phi} = 0.$$

where A, C are integration constants, ν is adiabatic index of the fluid.

Formation of Autonomous System

To study the system qualitatively, we will formulate an autonomous system from the above system.

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

Dimensionless Variables

To formulate autonomous system, we employ the dimensionless variables

$$x = \sqrt{\frac{k}{6}}\frac{\dot{\phi}}{H} \text{ and } y = \sqrt{\frac{k}{3}}\frac{\sqrt{V(\phi)}}{H}$$

and write the fractional energy densities as

$$\Omega_m = \frac{kAa^{-3}}{3H^2} \text{ and } \Omega_b = \frac{kCa^{-3\nu}}{3H^2}.$$

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

Autonomous System

We get the autonomous system as

$$\frac{dx}{dN} = 3x(x^2 - 1 + \frac{\Omega_m}{2} + \frac{\nu\Omega_b}{2}) - \sqrt{\frac{3}{2k}}\frac{V'(\phi)}{V(\phi)}y^2$$
$$\frac{dy}{dN} = y[3(x^2 + \frac{\nu\Omega_b}{2} + \frac{\Omega_m}{2}) + \sqrt{\frac{3}{2k}}\frac{V'(\phi)}{V(\phi)}x]$$
$$\frac{d\Omega_m}{dN} = -3\Omega_m[1 - \Omega_m - 2x^2 - \nu\Omega_b]$$

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

Jacobian Matrix

Consider linear perturbations about the critical point (x_c, y_c, Ω_{mc}). Also we assume $\sqrt{\frac{3}{2k}}\frac{V'(\phi)}{V(\phi)} = \text{constant} = \alpha$ (say). Then, We get the matrix as

$$\begin{bmatrix} 9x_c^2(1-\frac{\nu}{2}) - 3(1-\frac{\Omega_{mc}}{2}) + 3(1-\Omega_{mc})\frac{\nu}{2} & -3\nu y_c x_c - 2\alpha y_c & \frac{3x_c}{2}(1-\nu) \\ 3y_c x_c(2-\nu) + \alpha y_c & 3x_c^2(1-\frac{\nu}{2}) + \frac{3\nu}{2} + \frac{3\Omega_{mc}}{2}(1-\nu) + \alpha x_c - \frac{9}{2}\nu y_c^2 & \frac{3x_c}{2}(1-\nu) \\ -6x_c\Omega_{mc}(\nu-2) & -6\nu y_c \Omega_{mc} & 6\Omega_{mc}(1-\nu) + 3x_c^2(2-\nu) - 3\nu y_c^2 - 3(1-\nu) \end{bmatrix}$$

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

TABLE 1: Equilibrium Points and Eigen Values

Equilibrium Point	x	y	Ω_m	Eigen Values
C_1	0	0	0	$-3(1-\nu), \frac{3\nu}{2}, 3(\frac{\nu}{2}-1)$
C_2	1	0	0	$6(1-\frac{\nu}{2}), 3+\alpha, 3$
C_3	-1	0	0	$6(1-\frac{\nu}{2}), 3-\alpha, 3$
C_4	0	0	1	$-\frac{3}{2}, \frac{3}{2}, 3(1-\nu)$

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

2D Autonomous System

Considering $\Omega_m = 0.30$, we can reduce the previous 3D system to 2D system as

$$\frac{dx}{dN} = 3x[x^2 - 1 + \frac{\Omega_m}{2} + \frac{\nu}{2}(1 - \Omega_m - x^2 - y^2)] - \alpha y^2,$$
$$\frac{dy}{dN} = y[3x^2 + 3\frac{\nu}{2}(1 - \Omega_m - x^2 - y^2) + 3\frac{\Omega_m}{2}) + \alpha x].$$

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

TABLE 2: Critical Points for $\Omega_m = 0.3, \nu = 1.8, \alpha = -2$

Equilibrium Point	x	y	Nature
C_{2a}	0	0	Saddle
C_{3b}	0.52	0.70	Stable Node
C_{3c}	0.52	-0.70	Stable Node

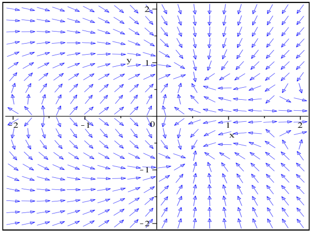


Figure: Phase portrait for the system, related to table 2

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

TABLE 3: Critical Points for $\Omega m = 0.32$, $\nu = 1.9$, $\alpha = 1.7$

Equilibrium Point	x	y	Nature
C_{2a}	0	0	Saddle
C_{2b}	-0.44	0.76	Stable Node
C_{2c}	-0.44	-0.76	Stable Node
C_{2d}	1.58	0	Unstable Node
C_{2e}	-1.48	0	Unstable

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

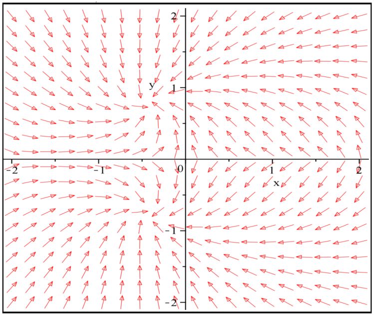


Figure: Phase portrait for the system, related to table 3

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

TABLE 4: Critical Points for $\Omega m = 0.2$, $\nu = 1.6$, $\alpha = 1.1$

Equilibrium Point	x	y	Nature
C_{2a}	0	0	Saddle
C_{4b}	-0.33	0.89	Stable Node
C_{4c}	-0.33	-0.89	Stable Node

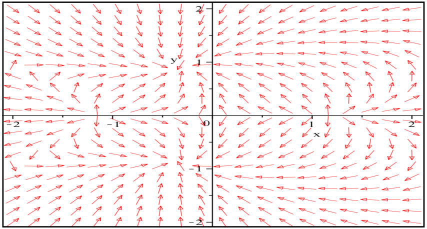


Figure: Phase portrait for the system, related to table 4

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

TABLE 5: Critical Points for $\Omega m = 0.22$, $\nu = 1.2$, $\alpha = 1.1$

Equilibrium Point	x	y	Nature
C_{2a}	0	0	Saddle
C_{5b}	-0.32	0.84	Stable Node
C_{5c}	-0.33	-0.89	Stable Node

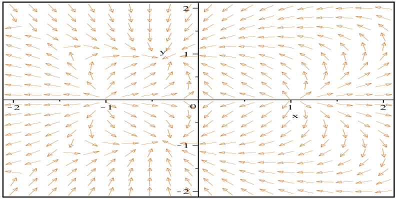


Figure: Phase portrait for the system, related to table 5

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

TABLE 6: Critical Points for $\Omega m = 0.24$, $\nu = 1.3$, $\alpha = 3$,

Equilibrium Point	x	y	Nature
C_{2a}	0	0	Saddle
C_{6b}	-0.58	0.47	Stable Node
C_{6c}	-0.58	-0.47	Stable Node

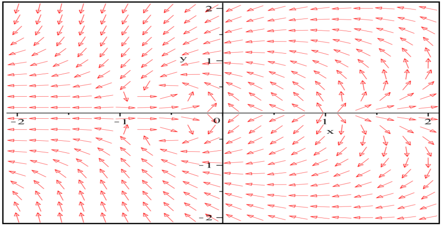


Figure: Phase portrait for the system, related to table 6

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

TABLE 7: Critical Points for $\Omega m = 0.24$, $\nu = 1.3$, $\alpha = 1.3$

Equilibrium Point	x	y	Nature
C_{2a}	0	0	Saddle
C_{7b}	-0.40	0.87	Stable Node
C_{7c}	-0.40	-0.87	Stable Node

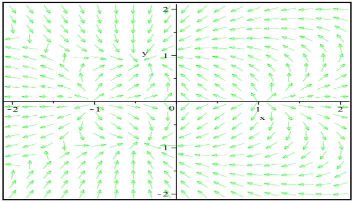


Figure: Phase portrait for the system, related to table 7

A Dynamical System Analysis of a Three Fluid Cosmological Model (Continued)

TABLE 8: Critical Points for $\Omega m = 0.2$, $\nu = 1.6$, $\alpha = -1.1$

Equilibrium Point	x	y	Nature
C_{2a}	0	0	Saddle
C_{4b}	0.33	0.89	Stable Node
C_{4c}	0.33	-0.89	Stable Node

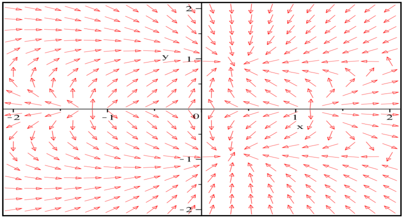


Figure: Phase portrait for the system, related to table 8

Dynamical System Analysis of Brane Induced Gravity with Tachyon Field

Paper Details

- **Title of the Paper:** Dynamical System Analysis of Brane Induced Gravity with Tachyon Field.
- **Author(s):** A. Ravanpak and G.F. Fadakar. (Department of Physics, Vali-e-Asr University, Rafsanjan, Iran)
- Class. Quantum Grav., 36, 235003 (2019)
  DOI 10.1088/1361-6382/ab4eec
- **Arxiv Number:** arXiv:1906.04465

Dynamical System Analysis of Brane Induced Gravity with Tachyon Field (Continued)

The Model

The Model is described by some basic equations. We have to transform them in the form of autonomous dynamical systems to study them properly.

The Friedmann Equation

The Friedmann equation in this case is as follows:

$$H^2 + \frac{H}{r_c} = \frac{1}{3M_p^2}(\rho_m + \rho_{tac}),$$

where r_c is the relation between the 4D and 5D Planck mass, H is the Hubble parameter, ρ_m is the matter energy density, ρ_{tac} denotes the energy density of the tachyon field and M_p is the 4D Planck mass.

Dynamical System Analysis of Brane Induced Gravity with Tachyon Field (Continued)

Energy Density and Pressure

The energy density and pressure of tachyon field are expressed as

$$\rho_{tac} = \frac{V(\phi)}{\sqrt{1-\dot{\phi}^2}}$$
$$P_{tac} = -V(\phi)\sqrt{1-\dot{\phi}^2}$$

in which ϕ and $V(\phi)$ are tachyon field trapped on the brane and the tachyon potential respectively and dot means derivative with respect to the cosmic time.

Dynamical System Analysis of Brane Induced Gravity with Tachyon Field (Continued)

Conservation Equations

The conservation equations are

$$\dot{\rho} + 3H\rho_m = 0$$
$$\dot{\rho}_{tac} + 3H(\rho_{tac} + P_{tac}) = 0$$

Equation of Motion

From the previous equations, we get the equation of motion of the tachyon field is

$$\frac{\ddot{\phi}}{1-\dot{\phi}^2} + 3H\dot{\phi} + \frac{V_\phi}{V} = 0$$

Dynamical System Analysis of Brane Induced Gravity with Tachyon Field (Continued)

Dimensionless Variables

We intend to describe this model as an autonomous system of ordinary differential equations. So, we introduce a set of dimensionless variables

$$x^2 = \frac{\rho_m}{3M_p^2(H^2+\frac{H}{r_c})},\ y^2 = \frac{V}{3M_p^2(H^2+\frac{H}{r_c})},\ d = \dot{\phi},\ z^2 = 1 + \frac{1}{Hr_c}$$

Dynamical System Analysis of Brane Induced Gravity with Tachyon Field (Continued)

Other Equations

After introducing the variables, the Friedmann constraint becomes

$$x^2 + \frac{y^2}{\sqrt{1-d^2}} = 1$$

and the Raychaudhury equation can be recast as

$$\frac{\dot{H}}{H} = \frac{-3z^2}{z^2+1}\left(x^2 + y^2\frac{d^2}{\sqrt{1-d^2}}\right)$$

Dynamical System Analysis of Brane Induced Gravity with Tachyon Field (Continued)

Autonomous System

From the previous equations, we get the autonomous system as

$$d' = -(3d - \sqrt{3}zy\lambda)(1 - d^2)$$
$$y' = -\frac{\sqrt{3}}{2}y^2dz\lambda + \frac{3}{2}y(1 - y^2\sqrt{1 - d^2})$$
$$z' = \frac{3}{2}\frac{z(z^2-1)}{z^2+1}(1 - y^2\sqrt{1 - d^2})$$

Here prime means derivative with respect to $\ln a$, and also

$$\lambda = -M_p V_\phi / V^{3/2}.$$

Dynamical System Analysis of Brane Induced Gravity with Tachyon Field (Continued)

Fixed Points and their properties when λ =constant

Point	(d, y, z)	Eigenvalues	Stability
P_1	(0, 0, 1)	(-3, 3/2, 3/2)	Saddle
$P_2^{\pm}$	($\pm$0.84, $\pm$0.73, 1)	(-3.7, -1.9, -1.9)	Stable
$P_3^{\pm}$	($\pm$1, 0, 1)	(6, 3/2, 3/2)	Unstable

Dynamical System Analysis of Brane Induced Gravity with Tachyon Field (Continued)

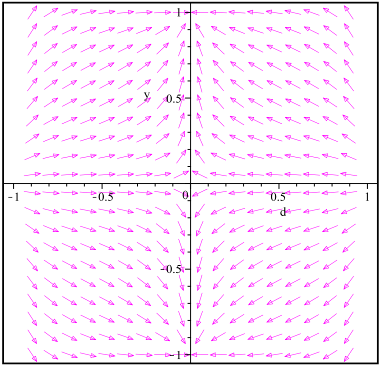


Figure: The 2D phase space (y, d) for $\lambda = 0$, treating z as constant (of any value, in this case)

Dynamical System Analysis of Brane Induced Gravity with Tachyon Field (Continued)

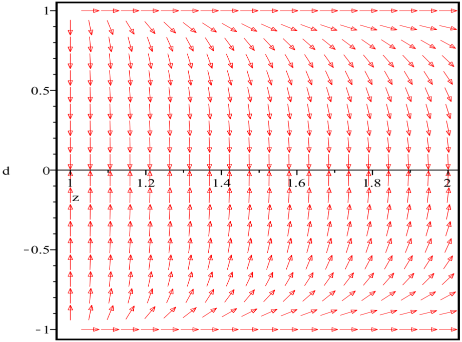


Figure: The 2D phase space (d, z), for $\lambda = 0$ and $y = \pm 1$.

Dynamical System Analysis of Brane Induced Gravity with Tachyon Field (Continued)

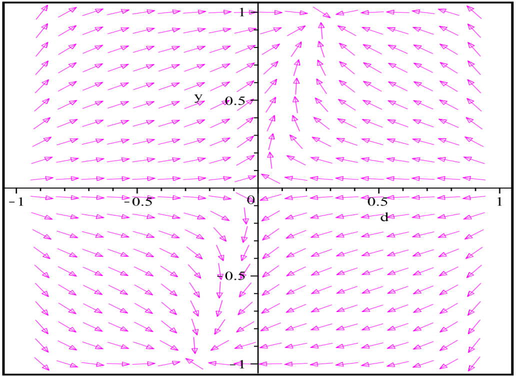


Figure: The 2D phase space (y, d), for $\lambda = 0.1$ and $z = 5$.

Dynamical System Analysis of Brane Induced Gravity with Tachyon Field (Continued)

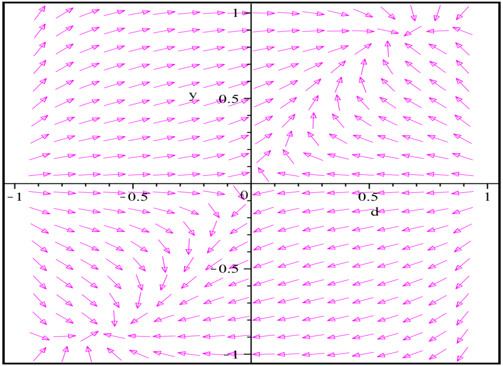


Figure: The 2D phase space (y, d), for $\lambda = 1.4$ and $z = 0.9$.

Dynamical System Analysis of Brane Induced Gravity with Tachyon Field (Continued)

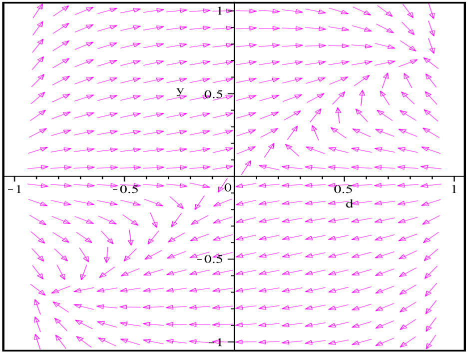


Figure: The 2D phase space (y, d), for $\lambda = 2.4$ and $z = 0.9$.

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Thank You!